# Theory of Computation and Automata

Assignment 1

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**Problem 1.** List the first 4 strings of the following languages:

a) $L = \{bbwbb : w \in \{a, b\}^*\}$
b) $L = \{(bbb)^n : n \geq 0\}$
c) $L = \{bbb^n : n \geq 0\}$
d) $L = \{b^n a^n : n \geq 1\}$
e) $L = \{w_1 aaw_2 : w_1 \in \{a, b\}^* \text{ and } w_2 \in \{c, d\}^*\}$
f) $L = \{a^{2n} ba^m : n, m \geq 0\}$
g) $L = \{w : w \in \Sigma^* and \sum = \{a, b\} and |w| \bmod 3 = 0\}$
h) $L = \{w^R w : w \in \Sigma^+ and \Sigma = \{1,2\}\}$
i) $L = \{w : w \in \Sigma^* and \sum = \{a, b\} and N_a(w) \bmod 2 = 0 and N_b(w) \geq 0\}$

**Problem 2.** (Operations on languages)

let $L_1 = \{00, 10\}, L_2 = \{\lambda, 0, 010\}$, $L_3 = \{0(10)^n : n \geq 0\}$ , $L_4 = \{0^n 1^m : n \geq 0, m \geq 1\}$ and $L_5 = \{w : w \in \{0,1\}^* \text{ and } N_0(w) = N_1(w)\}$,where alphabet $\sum = \{0, 1\}$,

a) List the strings in $\Sigma^3$.(concatenation 3 times)
b) List the strings in $L_1 L_2$. (concatenation)
c) List the first 5 string in $L_2^*$. (Kleene star)
d) List the strings in $L_2 \cap L_3$.
e) List the strings in $L_4 \cap L_5$.
f) List the strings in $L_4 \cup L_5$.
g) Write $L_3^R$ *in set builder notation(pattern/rule)*.

**Problem 3.** Write the following languages in (Set builder /Pattern/Rule) notation?

a) The set of strings over {a,b,c} that contain the substring **bb**.
b) The set of strings over {a,b,c} that end with **bb**.
c) The set of strings over {a,b,c} that begin with **bb**.
d) The set of strings over {a,b,c} in which the number of **b's** in the string is odd.
e) The set of strings over {a,b,c} in which the length of the string is **even**.
f) The set of strings over {a,b,c} in which the total number of a's ,b's and c's is greater than **two**.
g) The set of strings over {a,b,c} in which the total number of a's and c's is **two**.